

Grade 11 Mathematics

Exponents and Surds & Equations and Inequalities

Task: Task 8

Term: Term 3 2026

Time: 60 minutes

Total Marks: 46

Examiner: AE Killer

Moderator: D Piters

QUESTION 1

1.1 Answer the following questions without using a calculator.

1.1.1 Write 32 as a power of 2, and hence write down the value of x if $2^x = 32$. (1)

1.1.2 Solve for x : $2^{x+3} + 2^{x+1} = 320$ (4)

1.2 Given: $x = \sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$, where the pattern of nested square roots continues indefinitely.

1.2.1 By noting that the expression under the outermost square root, $12 + \sqrt{12 + \sqrt{12 + \dots}}$, contains the same infinite expression as x , write x^2 in terms of x . (2)

1.2.2 Hence, determine the value of x . (4)

1.3 Solve for x , without using a calculator: $\sqrt{2^x + 5} + 1 = 2^x$ (6)

1.4 Rectangle ABCD has side $AB = \sqrt{3} + 1$ units and diagonal $AC = 2\sqrt{2}$ units. Without using a calculator, determine the area of rectangle ABCD. Show all your working. (4)

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QUESTION 2

2.1 Given the inequality $(x+1)(x-4) \geq 6$

2.1.1 Show that the inequality can be written as $x^2 - 3x - 10 \geq 0$. (1)

2.1.2 Hence, solve for x : $(x+1)(x-4) \geq 6$ (4)

2.2 Determine the value(s) of k for which the equation $x^2 + (k-1)x + k + 2 = 0$ (5)

has equal roots.

2.3 Solve for x : $x(3-2x) \leq -9$ (5)

2.4 A farmer builds a rectangular chicken run against the straight wall of a barn. The wall forms one side of the run, so fencing is only needed for the other three sides. The farmer has exactly 50 m of fencing and uses all of it. The barn wall is 25 m long. Let x be the length, in metres, of each of the two sides of the run that are perpendicular to the wall.

2.4.1 Write down an expression, in terms of x , for the length of the side of the run that is parallel to the wall. (1)

2.4.2 The area enclosed by the run must be exactly 300 m². Determine the dimensions of the run. Take into account all the information given, and justify any value you reject. (4)

2.5 Given the equation $2x^2 - 5x - 1 = 0$

2.5.1 Solve for x : $2x^2 - 5x - 1 = 0$ (correct to TWO decimal places) (3)

2.5.2 Hence, solve for x : $2(x-3)^2 - 5(x-3) - 1 = 0$ (correct to TWO decimal places) (2)

[25]

TOTAL [46]

MARKING GUIDELINE

QUESTION 1		
1.1.1	$32=2^5$, so $x=5$	✓ answer (1)
1.1.2	$2^x(2^3+2)=320$	✓ factorise common power
	$10 \cdot 2^x=320$ so $2^x=32$	✓ simplify coefficient
	$2^x=2^5$	✓ same base
	$x=5$	✓ answer (4)
1.2.1	$x^2=12+\sqrt{12+\sqrt{12+\dots}}$	✓ square both sides
	$x^2=12+x$	✓ answer (2)
1.2.2	$x^2-x-12=0$	✓ standard form
	$(x-4)(x+3)=0$	✓ factors
	$x=4$ or $x=-3$	✓ roots
	$x=4$ (reject $x=-3$ since a square root cannot be negative)	✓ rejection of negative root
	$x=4$	(4)

1.3	$\sqrt{2^x+5}=2^x-1$	✓ isolate the surd
	$2^x+5=2^{2x}-2\cdot 2^x+1$	✓ square both sides
	Let $k=2^x$: $k^2-3k-4=0$	✓ standard form
	$(k-4)(k+1)=0$	✓ factors
	$2^x=-1$ (NA, since $2^x>0$)	✓ rejection of extraneous root
	$2^x=4=2^2$, so $x=2$	✓ answer (6)
1.4	$BC^2=AC^2-AB^2=(2\sqrt{2})^2-(\sqrt{3+1})^2$	✓ Pythagoras substitution
	$BC^2=8-(4+2\sqrt{3})=4-2\sqrt{3}$	✓ expand and simplify
	$4-2\sqrt{3}=(\sqrt{3}-1)^2$, so $BC=\sqrt{3}-1$	✓ length of BC
	Area = $(\sqrt{3+1})(\sqrt{3}-1)=3-1=2$ square units	✓ area (4)

QUESTION 2		
2.1.1	$x^2-4x+x-4-6\geq 0$	
	$x^2-3x-10\geq 0$	✓ expansion to standard form (1)

2.1.2	$(x-5)(x+2) \geq 0$	✓ factors
	Critical values: $x = -2$ and $x = 5$	✓ critical values
	$x \leq -2$ or $x \geq 5$	✓ ✓ both parts of answer (4)
2.2	$\Delta = (k-1)^2 - 4(1)(k+2)$	✓ substitution into discriminant
	$\Delta = k^2 - 6k - 7$	✓ simplified discriminant
	$k^2 - 6k - 7 = 0$	✓ condition $\Delta = 0$
	$(k-7)(k+1) = 0$	✓ factors
	$k = 7$ or $k = -1$	✓ answer (5)
2.3	$-2x^2 + 3x + 9 \leq 0$	✓ expand and standard form
	$2x^2 - 3x - 9 \geq 0$	✓ divide by -1, reverse sign
	$(2x+3)(x-3) \geq 0$	✓ factors
	Critical values: $x = -\frac{3}{2}$ or $x = 3$	✓ critical values
	$x \leq -\frac{3}{2}$ or $x \geq 3$	✓ answer, both intervals (5)

2.4.1	$50 - 2x$ m	✓ expression (1)
2.4.2	$x(50 - 2x) = 300$	✓ setup
	$2x^2 - 50x + 300 = 0$, so $x^2 - 25x + 150 = 0$	✓ standard form
	$(x - 10)(x - 15) = 0$, so $x = 10$ or $x = 15$	✓ both values
	If $x = 10$, the parallel side is 30 m, which is longer than the 25 m wall, so reject it. The run is 15 m by 20 m, with the 20 m side along the wall.	✓ rejection and answer (4)
2.5.1	$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-1)}}{2(2)}$	✓ substitution
	$x = \frac{5 \pm \sqrt{33}}{4}$	
	$x = 2,69$ or $x = -0,19$	✓ ✓ both answers (3)
2.5.2	$x - 3 = 2,69$ or $x - 3 = -0,19$	✓ link to previous answers
	$x = 5,69$ or $x = 2,81$	✓ both answers (2)

TOTAL [46]

COGNITIVE LEVEL ANALYSIS GRID

Weighting (Prescribed)			20%			35%			30%			15%			100%
Actual %			4%			50%			28%			17%			100%
Topic	Q	Sub-topic	Knowledge			Routine			Complex			Problem Solving			Total
			E	M	D	E	M	D	E	M	D	E	M	D	
Exponents and Surds	1.1.1	write number as prime power	1												1
	1.1.2	exponential eq, factor common power					4								4
	1.2.1	express infinite surd recursively					2								2
	1.2.2	solve quadratic, reject negative root				4									4
	1.3	surd with exponential, reject root								6					6
	1.4	surd rectangle area via pythagoras											4		4
Equations and Inequalities	2.1.1	expand to standard form	1												1
	2.1.2	quadratic inequality after expanding					4								4
	2.2	equal roots, find parameter					5								5
	2.3	quadratic inequality, rearrange first								5					5
	2.4.1	express side in terms of x				1									1
	2.4.2	area word problem, reject root											4		4
	2.5.1	quadratic formula to 2 d.p.					3								3
	2.5.2	hence, substitution into roots								2					2
TOTAL			2	0	0	5	18	0	0	7	6	0	0	8	46
%			4%			50%			28%			17%			100%

E = Easy, M = Medium, D = Difficult.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} ; r \neq 1$$

$$S_\infty = \frac{a}{1 - r} ; -1 < r < 1$$

$$F = \frac{x[(1+i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1+i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x-a)^2 + (y-b)^2 = r^2$$

$$\text{In } \triangle ABC: \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2}ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2\sin^2 \alpha \\ 2\cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2\sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$